

Name: Parthiv Patel

10

TRANSPORT PHENOMENA II, CHE 311

QUIZ 4

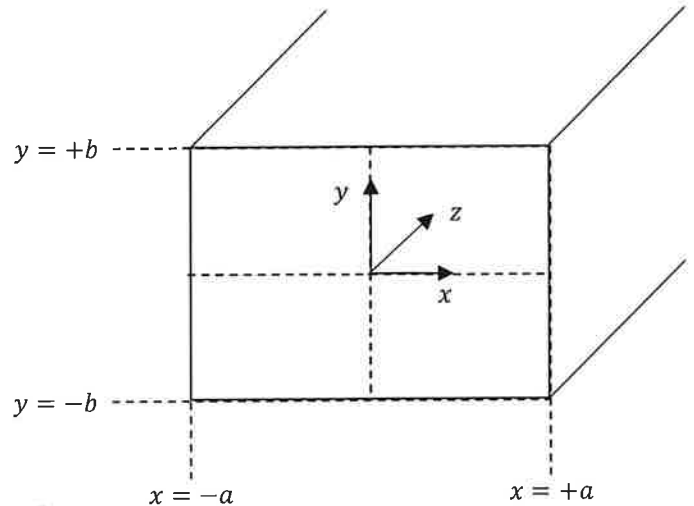
CLOSED BOOK, TIME: 15 MINUTES

A fluid of constant density ρ and viscosity μ is flowing in a rectangular duct of length L . The modified pressure at the inlet and the outlet of the duct are $\mathcal{P}_0$ and $\mathcal{P}_L$.

- 1- Write down the partial differential equation (PDE) that describes the steady-state velocity profile of the fluid.

$$V_z = \quad V_x = 0 \\ V_y = 0$$

$$\rho \left(\frac{\partial V_z}{\partial t} \right) = - \frac{\partial \mathcal{P}}{\partial z} + \mu \left[\frac{\partial^2 V_z}{\partial z^2} \right] + \rho g_z$$



in z direction
gravity is zero.

+ 10

- 2- Write down the four boundary conditions required to solve the above PDE.

$\frac{\partial V_z}{\partial t} = 0$ when flow is zero in z-direction
we can integrate PDE to find C_1 & C_2

$$V_z (= V_z(t \rightarrow 0, z=0)) = 0$$

$$V_x = 0$$

$$V_y = 0$$

$$0 = - \frac{\partial \mathcal{P}}{\partial z} + \mu \left[\frac{\partial^2 V_z}{\partial z^2} \right]$$

$\rho g_z = 0$ in z direction gravity is zero.

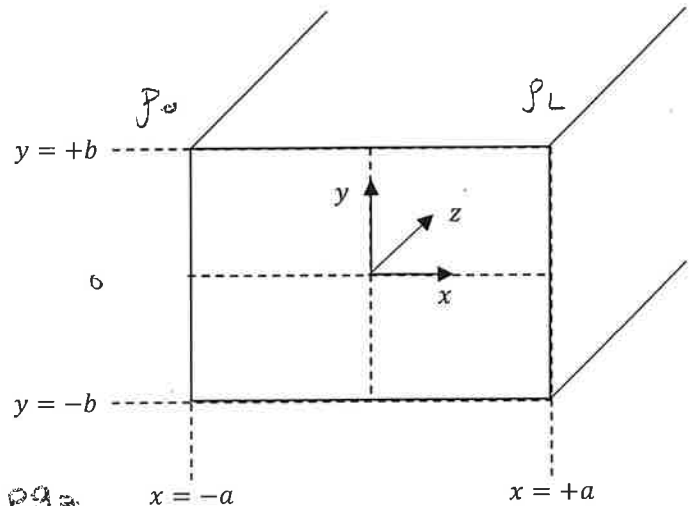
-40

TRANSPORT PHENOMENA II, CHE 311

QUIZ 4

CLOSED BOOK, TIME: 15 MINUTES

A fluid of constant density ρ and viscosity μ is flowing in a rectangular duct of length L . The modified pressure at the inlet and the outlet of the duct are P_0 and P_L .



- 1- Write down the partial differential equation (PDE) that describes the steady-state velocity profile of the fluid.

$v_z = v_z(y, t)$
steady state

$v_z = v_z(y)$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \rho g_z$$

$$0 = -\frac{dP}{dz} + \mu \left[\frac{d^2 v_z}{dy^2} \right] + \rho g_z$$

$$\mu \frac{d^2 v_z}{dy^2} = \frac{P_0 - P_L}{L}$$

-15

- 2- Write down the four boundary conditions required to solve the above PDE.

$$v_z(b, t) = 0$$

$$v_z(-b, t) = 0$$

$$v_z(-a, t) = 0$$

$$v_z(a, t) = 0$$

-15

A fluid of constant density ρ and viscosity μ is flowing in a rectangular duct of length L . The modified pressure at the inlet and the outlet of the duct are P_0 and P_L .

$$v_z(x, y, z)$$

- 1- Write down the partial differential equation (PDE) that describes the steady-state velocity profile of the fluid.

$$v_z(x, y)$$

$$\rho(-\Delta + 0 + 0, t_0)$$

$$= -\frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} \right] + 0$$

$$\Delta = F - I$$

$$0 = -\frac{\partial P}{\partial z} + \mu \left[\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} \right]$$

$$* \frac{\partial P}{\partial z} = \frac{P_L - P_0}{L}$$

$$\frac{P_L - P_0}{L} = \mu \frac{\partial^2 v_z}{\partial x^2} + \mu \frac{\partial^2 v_z}{\partial y^2}$$

- 2- Write down the four boundary conditions required to solve the above PDE.

BC ①: $v_z(x=a, y) = 0$ ③ $v_z(x, y=b) = 0$
 ② $v_z(x=-a, y) = 0$ ④ $v_z(x, y=-b) = 0$

